BRIEFING

European Parliament

Powering the EU's future: Strengthening the battery industry

SUMMARY

Batteries, widely used in the transport and energy sectors, are central to the global energy system. They will be key to the EU's clean energy transition, industrial future and strategic autonomy. Boosting the industrial base for battery production is therefore a key task for the EU.

While the EU battery sector enjoys strong support for its research and development activities, it also faces significant challenges such as dependencies on third countries and high energy and labour costs. Developing battery recycling or implementing more sustainable value chains could help address some of these challenges. However, significant threats, such as competition from countries with lower environmental standards, may hinder progress and require strategic action.

Projections around battery manufacturing in the EU remain highly uncertain. Many reports claim that the EU is on track to meet its future battery needs, yet also highlight significant risks that could prevent this from happening. Factors such as rising energy and labour costs, incentives offered by third countries, slower-than-expected market developments, or difficult access to critical raw materials, could undermine the sector's competitiveness. Mastering the complexities of battery manufacturing technologies remains another major challenge. Delays or cancellations of gigafactory projects have already been announced across Europe. The recent collapse of Northvolt, once hailed as Europe's flagship home-grown battery manufacturer, has raised serious concerns about the future of batteries 'made in Europe'.

The EU has long recognised batteries as one of its strategic technological sectors. To make its battery supply chains secure, resilient and sustainable, the EU uses three approaches. First, it seeks to inject strategic impetus into the sector, using its convening power to improve cooperation among stakeholders. Second, it is working on a comprehensive regulatory framework. Third, it provides the sector with funding. The development of the battery sector provides an instructive case study for shaping an effective EU industrial policy.

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Introduction

Battery technologies are at the heart of the global energy system. Electrification – the process of replacing technologies and systems that rely on fossil fuels with those powered by electricity, often sourced from renewable energy – has enhanced their importance. These technologies support the roll-out of clean transportation and stabilise the power grid. Besides, they are vital for decarbonising economies, strengthening energy security and boosting competitiveness in the global energy economy. Access to batteries is particularly important for ensuring the competitiveness of the automotive sector, as batteries are the most critical component of electric vehicles and represent around 25-40 % of the total cost of an electric vehicle. This explains why countries all over the world have been stepping up their efforts to boost their battery manufacturing capacity. For example, having prioritised the manufacturing of electric vehicles (EVs) and their batteries in the early 2000s, China invested US$230 billion in the sector from 2009 to 2023. Similarly, through the adoption of its Inflation Reduction Act in 2022, the United States introduced incentives for domestic battery production. This has resulted in investment in batteries and critical minerals refining more than tripling, with battery manufacturing investment reaching US$40.9 billion.

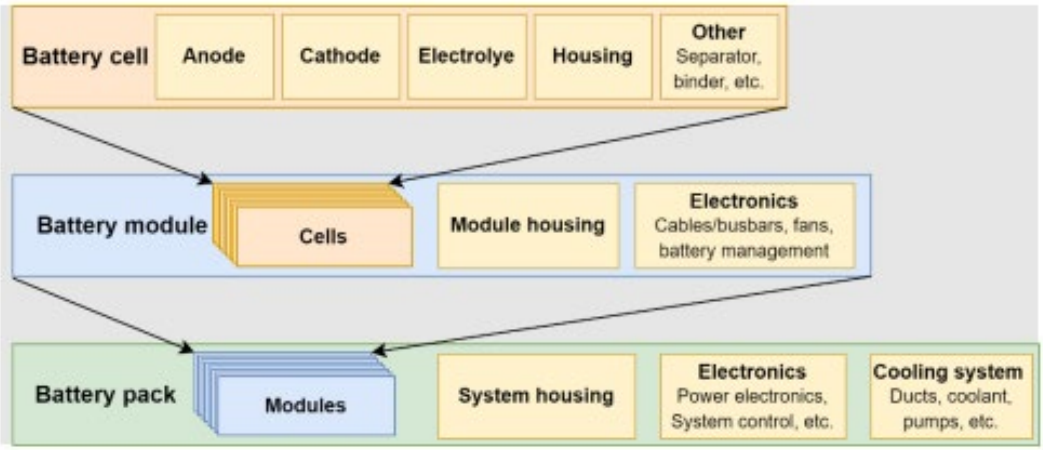
Since 2018, global investment in EV batteries and in battery storage has increased eightfold and fivefold, respectively, reaching a total of US$150 billion in 2023. Of this amount, 75 % was for EV batteries. In recent years, the global battery market has been growing dramatically: between 2021 and 2023, global battery manufacturing capacity increased from 1.1 terawatt hours (TWh) to 2.5 TWh. Announced expansions could lead to a manufacturing capacity of 9.3 TWh in 2030. EVs account for over 90 % of battery use, with lithium-ion batteries being the most used technology.

The battery supply chain is highly concentrated both as regards the manufacturing of technologies and the materials on which these technologies rely. In the renewed EU industrial strategy, the European Commission highlights the strategic dependencies and related risks for the EU in the batteries sector. In his report of October 2024, former Finnish President, Sauli Niinistö, reiterates warnings that 'supply chain dependencies ... and new clean technologies are leveraged by competing and malicious global powers to create the potential for weaponisation and build up leverage to be used as part of coercive strategies'. The report therefore identifies boosting the EU's industrial base for battery production as a key task. However, the EU's ambition to become a global battery powerhouse faces many challenges. In particular, building batteries in 'gigafactories' (large-scale battery production plants) requires mastering complex, rapidly evolving technologies. Recent developments in the sector have had a chilling effect on industry players: in November 2024, Northvolt, once hailed as Europe's flagship home-grown battery manufacturer and the continent's best-funded start-up, filed for bankruptcy, and its CEO resigned. Over its lifetime, the company had secured US$15 billion in funding, including US$5 billion in grants and loans from Canada, Germany, Germany, Poland, Sweden and the EU. It also attracted significant backing from US investors and major European car makers. The company's collapse has been attributed to a combination of factors, including poor management practices, substandard safety protocols, over-reliance on Chinese and South Korean machinery and workers, and unresolved technological issues. Amidst these challenging circumstances, can the EU achieve its ambitions for battery production?

Battery technologies: Lithium-ion and beyond

A rechargeable battery is an energy storage device that can convert chemical energy into electrical energy and vice versa. The basic unit of a battery is called a battery **cell**. A battery cell typically contains two electrodes (a **cathode**, carrying a positive electrical charge, and an **anode**, carrying a negative electrical charge) which are in close contact with an **electrolyte** (a chemical solution that allows ions (charged particles) to move between the two electrodes), and a separator (a physical barrier between the cathode and the anode). Battery cells are contained in battery **modules**, which in their turn are contained in a battery **pack** (see Figure 1).

Figure 1 – What is inside a typical battery



Source: Joint Research Centre's Science for policy report, 2023.

Cells typically account for 70 % to 85 % of the total battery weight. They contain a number of minerals in the cathode (e.g. lithium, nickel, cobalt and manganese), anode (e.g. graphite) and current collector (e.g. copper). The modules and pack components consist mostly of aluminium, steel, coolants and electronic parts. Battery production is very energy- and time consuming, resulting in high manufacturing costs. Materials account for about 50-70 % of a battery's final cost. A battery's chemistry refers to the materials used to produce the battery, and batteries are commonly classified according to the chemistry of their cathode. The different combinations of materials result in different battery characteristics.

The inventors of lithium-ion (Li-ion) batteries were awarded the Nobel Prize in chemistry in 2019. Over the past decade, these batteries have overtaken the battery sector thanks to a 90 % cost reduction since 2010, a higher energy density (density, measured in watt-hours per kilogram (Wh/kg), is the amount of energy a battery can store with respect to its mass) and a longer lifetime. They have been used in a wide range of sectors, from consumer electronics, utility-scale/stationary power storage systems, to EV systems. This allows them to be stacked into lighter (lithium is the lightest metal) and more compact battery packs than those made from other materials. However, while energy density is of key importance for EV batteries, it is less important for battery storage, leading to a significant shift towards lithium iron phosphate (LFP) batteries in this sector.

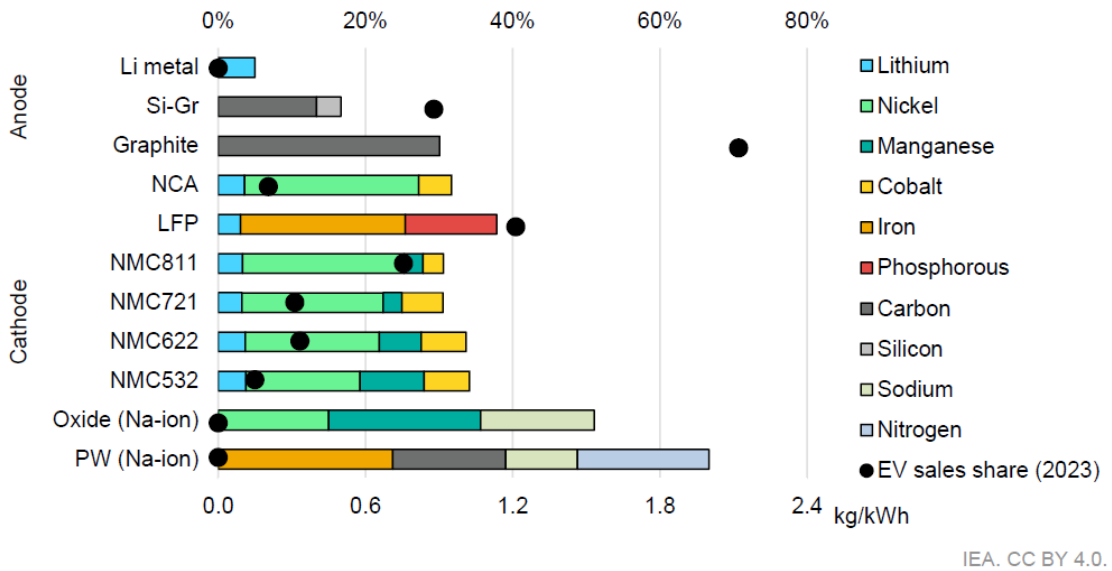
Despite what its name suggests, a Li-ion battery's chemistry is not exclusively composed of lithium; elements such as nickel, cobalt and manganese are essential components of its cathode (see Figure 2). Today, the most commonly used cathode chemistries are:

- lithium nickel manganese cobalt oxide (NMC).
- lithium nickel cobalt aluminium oxide (NCA).
- lithium iron phosphate (LFP).
- lithium cobalt oxide (LCO).

Yet another cathode chemistry classification system relies on the different combinations and proportions of the materials used (a.k.a. 'stoichiometric ratios'). For instance, NMC 811 batteries have a higher nickel content than NMC 532 batteries. Chemistries can be adapted to mineral availability and price. NMC and NCA batteries have a strong presence among the batteries used in EVs because their high nickel content leads to higher energy densities. Currently, LFP batteries are more than 20 % cheaper than NMC ones but have a lower energy density (-20-30 %). Thanks to LFP batteries' lower price and longer lifespan, they have been the leading type of batteries, in terms of their chemistry, in new EVs in China since 2021. More recently, European and US car makers have started making greater use of them as well. The market share of LFP batteries grew to 40 % of EV

sales and 80 % of new battery storage in 2023. The most common materials used for the LFP **anode** is graphite. Some manufacturers also use lithium titanate and efforts to use silicon are underway.

Figure 2 – Chemistry and mineral content of Li-ion batteries' anodes and cathodes (2023)



Notes: EV = Electric vehicle; Li metal = lithium metal anode; Si-Gr = silicon-graphite anode; Graphite = pure graphite anode; Na-ion = sodium-ion; LFP = lithium iron phosphate; NMC = lithium nickel manganese cobalt oxide; NCA = lithium nickel cobalt aluminium oxide; PW = Prussian white; Oxide (Na-ion) = $Na_{0.83}Ni_{0.32}Mn_{0.47}Mg_{0.1}Ti_{0.12}O_2$, for which magnesium (Mg) (0.06 kg/kWh) and titanium (Ti) (0.14 kg/kWh) are not displayed for simplicity. Other chemistries for sodium-ion batteries are being developed as well, most of them based on layered oxide. The oxygen content of the different chemistries is not displayed for simplicity. Materials composing the battery casing and the electrolyte, separator and current collectors are excluded. Chemistries' vary in their composition based on demand.

Sources: IEA analysis based on data from EV Volumes, Bloomberg New Energy Finance, Argonne Laboratory, BatPaC, and Wood Mackenzie (2023), Sodium-ion batteries: disrupt and conquer?.

Source: International Energy Agency, 2024.

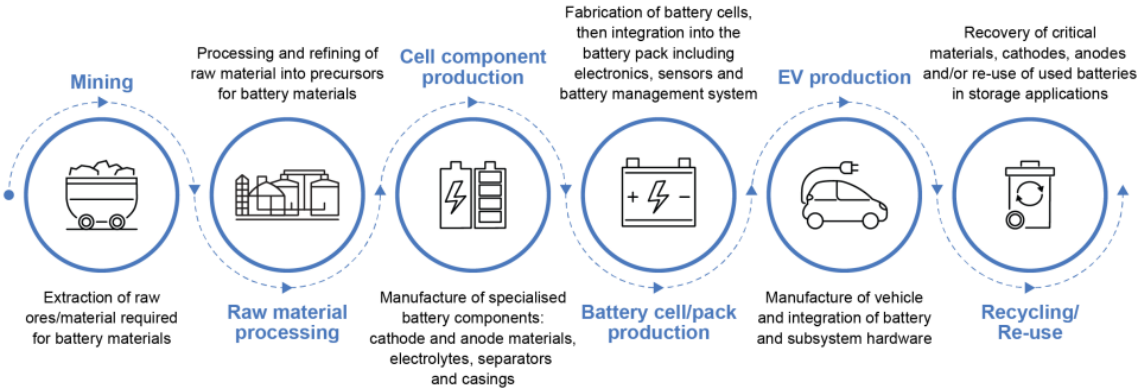
Research and innovation efforts are focused on enhancing the energy density, durability, safety and affordability of batteries, while minimising the environmental, social, and geopolitical impacts associated with sourcing their constituent materials. Further innovation in battery chemistries and manufacturing is projected to reduce global average lithium-ion battery costs by a further 40 % from 2023 to 2030. Beyond 2030, battery costs are likely to fall even more.

The role of other technologies is expected to increase. Sodium-ion batteries are a promising alternative, but the IEA considers that they are approximately 10 years away from widespread commercialisation. The sodium-ion technology does not rely on any critical raw materials. Solid-state batteries are on track to be commercially available and have the potential to bring massive performance gains. However, the actual merits of the new battery technologies and their potential remain uncertain until they have been commercially deployed.

Key features of the battery supply chain

The global battery supply chain, as shown in Figure 3, is complex. Every step in the process, from the extraction of mineral ores to the use of specific chemicals for manufacturing battery components in the final battery pack, is geographically highly concentrated.

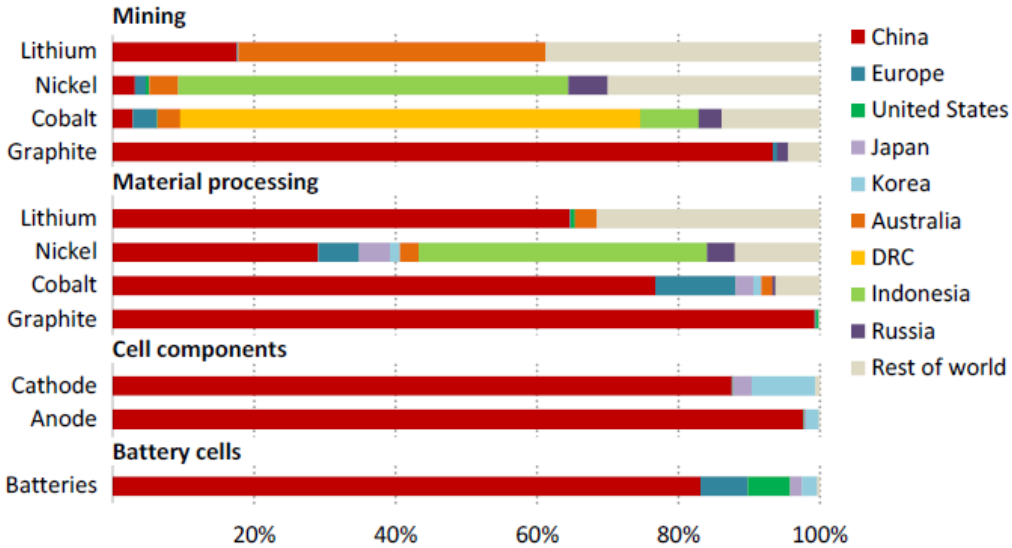
Figure 3 – Battery supply chain



Source: International Energy Agency, 2022.

Battery manufacturers are dependent on a small number of countries for the supply of raw materials and the extraction of many critical minerals (see Figure 4). China dominates the global battery supply chain.

Figure 4 – Geographical distribution of the battery supply chain (2023)



*DRC: Democratic Republic of Congo.

Source: International Energy Agency, 2023.

China accounts for more than 80 % of the manufacturing capacity for battery cells, followed by the United States and the EU with around 5 % each. More particularly, China accounts for nearly 90 % of the global capacity for cathode active materials and for over 97 % of the global capacity for anode active materials. For certain segments of the supply chain (such as cathodes and anodes), the EU's manufacturing capacity is extremely low.

As mentioned earlier, large-scale battery production plants are often referred to as 'gigafactories'. Battery production is a large volume, low profit-margin business, in which manufacturing optimisation, cost reduction and technological innovation play a key role in market competition. This offers a notable advantage to incumbent producers, as they can benefit from economies of scale and more efficient manufacturing while at the same time investing heavily in R&D. In Europe, Li-ion battery pack prices are around 163 €/kWh, which is 40 % higher than in China, where they are around 116 €/kWh. Anode and cathode active materials account for approximately 10 % and 50 % of the manufacturing costs for battery cells, respectively, and therefore make up the majority of overall battery manufacturing costs.

EU support for the battery industry

The EU has long recognised batteries as a strategic technology. The EU operates in the battery sector primarily in three ways. Firstly, by providing strategic impetus within the sector, using its convening power to improve cooperation among stakeholders and define an EU industrial strategy. Secondly, by implementing a comprehensive regulatory framework. Thirdly, by providing funding to the sector.

In 2007, the strategic energy and technology (SET) plan identified achieving a breakthrough in the cost efficiency of energy storage technologies as a key EU technological challenge on the path to decarbonising the EU economy. One of the ten priority actions under the 2015 revision of the SET plan was for the EU battery sector to achieve global competitiveness, enabling it to drive EU e-mobility forward. An important development in EU battery policy was the establishment of the European Battery Alliance (EBA) in 2017 – an industry-led cooperation platform tasked with spurring investment in the battery industry. Currently, the platform has 800 stakeholders.

The 2018 strategic action plan on batteries, based on proposals by the EBA, brought together a set of measures to support national, regional and industrial efforts across the value chain to boost innovation and enhance competitiveness in the EU battery industry. Member States' funding for battery producers is subject to the EU's state aid rules. So far, the European Commission has approved state aid for two important projects of common European interest (IPCEIs) in the area of batteries. The first one, approved in 2019, was worth €3.2 billion of state aid, and the second one (EuBatIn), approved in 2021, was worth €2.9 billion. The two projects are expected to leverage €14 billion in private investment. In March 2023, the Commission adopted a temporary crisis and transition framework to promote support measures in sectors that are key for the transition to a net-zero economy. It introduced specific measures enabling investment support for the manufacturing of strategic equipment such as batteries, which will be applicable until 31 December 2025.

The Green Deal industrial plan (GDIP) of 2023 presented a roadmap for boosting the EU's net-zero industry. Battery technologies are among the 19 strategic net-zero technologies within the scope of the Net-zero Industry Act (NZIA) adopted under the GDIP. The NZIA includes the non-legally binding objective of reaching an EU battery manufacturing capacity of at least 550 gigawatt hours (GWh) by 2030. Under the NZIA, all battery manufacturing projects recognised by a Member State as 'strategic' enjoy national-level priority status, shorter permitting timelines and streamlined procedures. Member States may also designate 'net-zero acceleration valleys': specific areas where net-zero industrial activities, including battery manufacturing, will be accelerated. The strategic technologies for Europe platform (STEP) aims to enhance value chains and provide support to the development and manufacturing of net-zero technologies (including batteries), as defined in the NZIA.

The Critical Raw Materials (CRMs) Act was also adopted under the GDIP. It includes measures to strengthen the EU's raw materials supply chains, monitor and mitigate supply risks and increase the sustainability of the CRMs consumed in the EU. It is expected to contribute to the EU battery industry's competitiveness. The EU's new Regulation on batteries and waste batteries introduces requirements on sustainability, safety, labelling, marking and information to allow the sale of

batteries within the EU. It also establishes minimum requirements for extended producer responsibility, the collection and treatment of waste batteries and reporting. Furthermore, it imposes battery due diligence obligations and lays down requirements for green public procurement when procuring batteries or products into which batteries are incorporated. The implementation of the regulation requires the adoption of key secondary legislation, such as a delegated act on the calculation and verification of the carbon footprint of EV batteries. The Euro 7 Regulation lays down provisions and requirements on battery durability (durability referring to the ability of a battery to maintain its required performance over time). Following its amendment in 2024, the EU Directive on industrial emissions (integrated pollution prevention and control) now includes provisions on large installations that manufacture batteries (as opposed to installations that only assemble batteries) with a production capacity of 15 000 tonnes of battery cells or more per year.

The EU has provided financial support to the sector, through channels such as Horizon Europe, the European Regional Development Fund and the Recovery and Resilience Facility. The European Court of Auditors (ECA) estimates that during the 2014–2020 period, the EU budget provided at least €1.7 billion in grants and loan guarantees to the EU battery industry. Furthermore, the Recovery and Resilience Facility has supported the construction of battery manufacturing plants.

In December 2024, the European Commission launched a €1 billion call for proposals for EV battery cell manufacturing (IF24 Battery), open until 24 April 2025. Grants will be funded under the EU Innovation Fund. In addition, the European Commission and the European Investment Bank (EIB) have presented a €200 million top-up (loan guarantee) from the EU Innovation Fund to the InvestEU programme to support investments in the EU battery manufacturing sector. The EIB has announced that it would invest a further €1.8 billion in the battery value chain.

The EU battery sector: State of play and projections

While there are some factors that reinforce the EU battery sector, such as strong support for R&D and policies that promote technologies spurring demand, it also faces significant challenges, including dependencies on third countries, competition with advanced battery producers, and high energy and labour costs (see Figure 5). Opportunities for the EU lie in the development of battery recycling technologies or more sustainable value chains that could help address some of these challenges. However, significant threats, such as continued dependence on external raw materials and geopolitical issues, may hinder progress and require strategic action.

In 2023, European-based battery production covered almost half of Europe's demand for batteries applied in EVs and energy storage systems. Production output reached nearly 100 GWh, led by LG Energy Solution in Poland (between 34 and 55 GWh), Samsung SDI (about 21–24 GWh) and SK On in Hungary (about 10–12 GWh), Northvolt in Sweden (up to about 6 GWh), BMZ (up to about 10 GWh) and CATL (5–8 GWh) in Germany. Korean companies account for three quarters of European production capacities.

Projections regarding future battery manufacturing in Europe, both by European and foreign companies, are highly uncertain. Numerous reports provide estimates suggesting that the EU is on track to meet its future battery needs, but they also highlight significant risks that could prevent these projections from being fully realised. The actual deployment of projected battery production capacity could be negatively affected by a range of factors, such as the time required for production facilities to start operating at full scale, decisions to cancel or postpone investments in response to incentives in other regions of the world, increases in energy or raw materials costs, rising energy and labour costs, or a lack of access to critical raw materials. Mastering the complexities of battery production technologies remains another major challenge. Delays or cancellations of gigafactory projects have already been announced across Europe.

Figure 5 – Competitiveness of the EU battery sector: A SWOT analysis

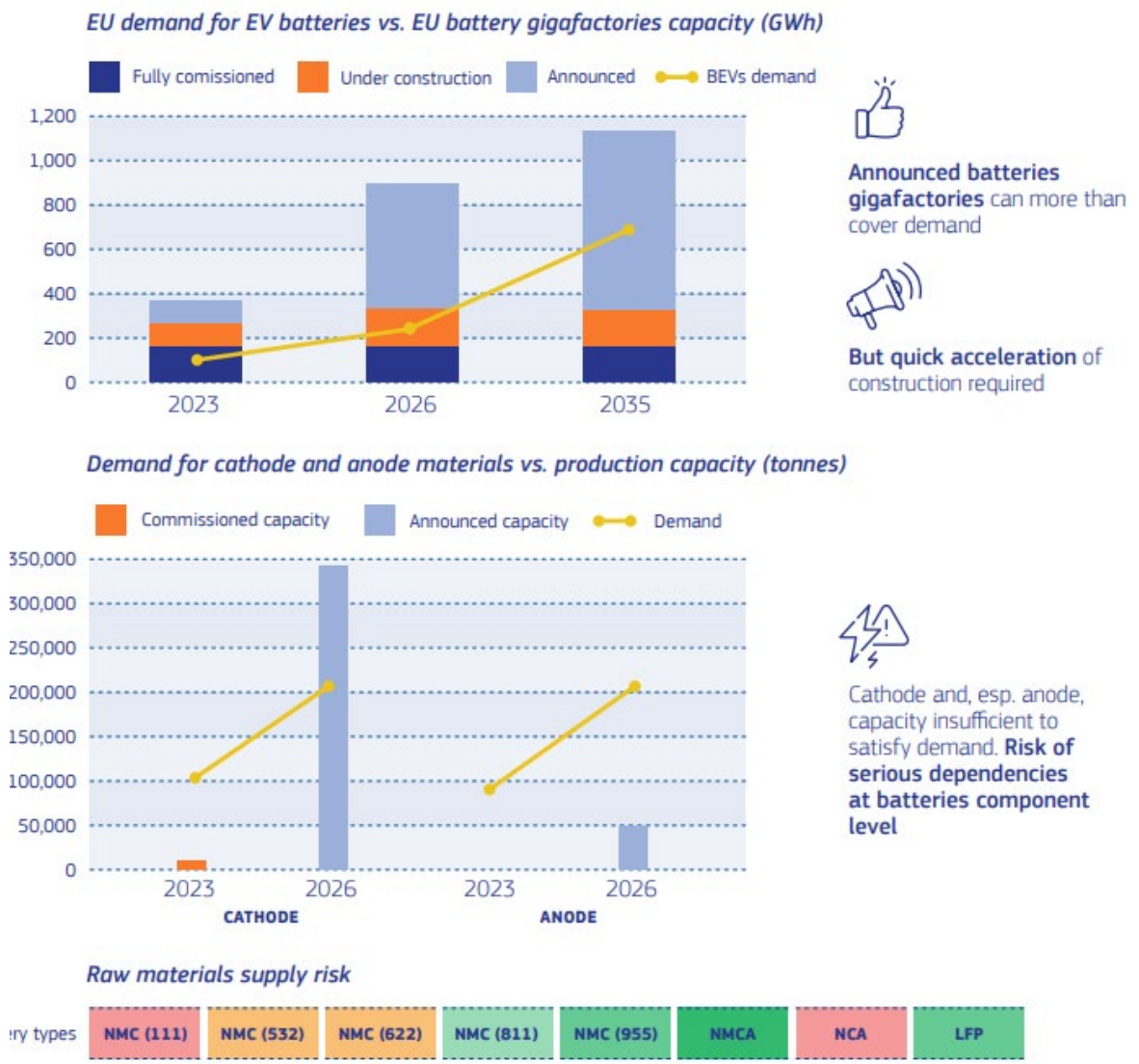
STRENGTHS	WEAKNESSES
• Strong automotive industry pushing demand; • Policies promoting technologies that support demand; • Strong EU support for R&D and deployment; • An overall strong R&D sector; • A well-educated work force; • General awareness of the need to mitigate climate change and maintain high environmental standards;	• Competition from well-established battery producers; • Dependency on third countries for raw materials supply and purification, and for production machinery; • High energy prices and labour costs; • Complex EU legislation and bureaucracy making the investment approval process lengthy; • Conflicting interests among Member States; • Shortage of workers specialised in battery manufacturing; • Regulations and standards not sufficiently developed; • Missing domestic production of cheap batteries and goods incorporating batteries; • Missing or weak links between the R&D sector and industry, particularly in eastern Europe; • High degree of business uncertainty and lack of trust in new technologies cripple producers who bring novel products to the market.
OPPORTUNITIES	**THREATS**
• Development of markets based on battery use; • Synergies with other value chains, e.g. hydrogen, other forms of energy storage and grid services; • Battery sector serving as an enabler of the wide deployment of renewable energy storage, energy grids stabilisation and reduction of energy costs; • Reduction of emissions from the use of fossil fuels; • Development of local value chains with reduced geopolitical risks for innovative, more sustainable or cheaper alternative battery chemistries; • EU has the potential to contribute to the development of missing international regulations and standards on batteries; • Battery recycling could partly cover demand for secondary raw materials, including CRMs;	• Continued dependence on external raw materials; • Growing demand for battery imports can worsen the EU's trade balance; • Emergence of third countries dominant in cheap battery technologies; • Risks from toxic materials in battery production and from improper management of use and recycling; • Supply of cheap batteries from countries with low environmental standards; • Geopolitical rivalry can trigger disinformation targeting EU citizens, aimed at limiting trust in battery technologies and promoting third countries as 'proven' suppliers of oil, gas, etc.; • Risk to EU competitiveness posed by third countries' subsidies policies and politically based decisions; • Customers might be unwilling to pay higher prices for EU-produced batteries (even if technically better);

Data source: Clean Energy Technology Observatory, Joint Research Centre, 2024.

In its 2023 report on progress on competitiveness of clean energy technologies, the Commission stated that the EU was on track to meet its projected demand (550 GWh in 2025 and 1000 GWh in 2030), as battery production in the EU was expected to reach 458 GWh by 2025 and 1083 GWh by 2030. However, the Commission cautioned that not all announced investments might ultimately materialise. As regards batteries for stationary energy storage in the EU (for energy grid or home storage), despite steady growth, their roll-out should accelerate to meet the forecast demand of 200 gigawatts (GW) by 2030.

At the 8th high-level meeting of the European Battery Alliance in May 2024, it was highlighted that a total of 30 gigafactory projects had been announced, with the potential to achieve a combined capacity of 1.3 TWh by 2030. Additionally, Europe's total installed battery manufacturing capacity had reached 167 GWh in 2023 (capacity utilisation rates of battery plants are not 100 %).

Figure 6 – Projections for the EU demand for EV batteries, cathode and anode materials, compared to existing capacity



Source: European Commission, Dashboard towards zero-emission vehicles, April 2024.

In June 2023, the ECA estimated that the EU's Li-ion battery cell production capacity may increase from 44 GWh in 2020 up to 1 200 GWh by 2030. Such domestic production could satisfy the expected EU demand by 2025 (400 GWh). The ECA pointed out that subsidiaries of non-EU companies currently own most of these manufacturing sites but EU-based companies are projected to gradually come into possession of a larger share of this production capacity, which could reach 56 % of overall EU production capacity in 2025.

In April 2024, the Commission reiterated what it had stated before: that the announced battery gigafactories can more than cover EU demand for EV batteries in 2035, but this requires accelerating the pace of construction (see Figure 6, above). However, at the level of battery components (cathode, and especially anode), there is a high risk of dependencies.

In May 2024, the European umbrella for NGOs working in the area of transport and the environment, Transport & Environment (T&E), estimated that Europe (more specifically the EU, Switzerland, Norway and the United Kingdom) could become self-sufficient in battery cells by 2026, and manufacture most of its demand for key components (cathodes) and materials such as lithium by 2030. Some 55 % of announced gigafactories capacities are from European companies. However, T&E pointed out that more than half of gigafactory plans in Europe remain at risk of either being delayed or cancelled.

The road ahead: Main issues and debate on possible new policy developments[1]

The ECA highlights the persistent challenge of securing access to raw materials for the EU's battery value chain, which remains heavily reliant on foreign suppliers and faces a potential shortage of battery raw materials. The ECA recommends that the Commission improve its monitoring of the supply chain and better target EU funding for the sector. Furthermore, it emphasises the importance of ensuring a level playing field for all participants in IPCEIs on batteries, particularly in accessing public financial support within their respective Member States.

The Draghi report on EU competitiveness strongly advocates establishing a comprehensive and coordinated CRMs strategy, as well as an EU industrial action plan for the automotive sector, both covering the whole value chain (including batteries). Draghi makes a number of proposals, including:

- incentivising the development of battery recycling standards;
- setting up net-zero acceleration valleys specialised in battery development and recycling under the NZIA;
- facilitating work permits (e.g. a green/blue card) for skilled professionals in the battery sector;
- supporting new automotive sector IPCEIs, for instance, ones focused on cost reduction through advancements in the technologies used in battery manufacturing;
- at the EU level, defining clean technologies, including batteries, as a strategic priority area in the 10th EU framework programme for research and innovation;
- creating a true single market for waste and recycling in the EU, which could boost battery recycling.

The report also pointed out that end-of-life lithium-ion batteries and intermediates of recycling, such as battery production waste and black mass, might be classified differently across Member States in the absence of EU rules on their classification as hazardous or non-hazardous waste.

The need to promote the recycling of batteries was also highlighted in the Commission's 2023 progress report on competitiveness of clean energy technologies as a way to increase the EU's presence in the early stages of the value chain and improve its security of supply. Another economic report, by Enrico Letta, also points to the need to make the single market more circular, in order to achieve material recovery economies of scale.

The European Battery Alliance points to the importance of including several key focus areas in an action plan for the 2024-2030 period aimed at strengthening the European battery value chain. Firstly, it is crucial to ensure fair competition with third countries while remaining open to free trade and building partnerships that enhance the competitiveness and resilience of the EU battery industry along the value chain. Secondly, there is a need to strengthen the demand for sustainable batteries produced in Europe (for instance, through public procurement). Thirdly, it is important to leverage the necessary financial instruments to support the scale-up of the battery industry. Fourthly, supporting industry-led R&D programmes is essential to maintain a level of technology sovereignty and accelerate time to market.

The main demands of the Association of European automotive and industrial battery manufacturers (Eurobat) focus on strengthening support for innovation in all mainstream battery technologies, ensuring a global level playing field for battery production and boosting circularity in the sector.

For Bruegel, the crisis at Northvolt should be considered a warning for the EU regarding the future direction of its industrial policy. It stresses systemic vulnerabilities for clean-tech in Europe, including the persistent reliance on foreign suppliers for critical inputs, the challenge of managing the rapid scaling-up of manufacturing capacity and the difficulty of competing with established battery producers in Asia. According to Bruegel, there are three lessons for the EU. Firstly, the EU should support multiple innovative players to build resilience, ensuring that failures in the innovation cycle do not jeopardise Europe's broader industrial strategy. Secondly, Europe should seek to build strategic partnerships with Asian companies, leveraging their knowledge and manufacturing efficiency while offering market access in return ('derisking by embracing'). Such partnerships should be governed by a solid regulatory framework to ensure European security interests – starting with cybersecurity (promoting a 'location over ownership' approach – with production within Europe, regardless of ownership). Thirdly, Europe now needs to design its own industrial policy approach, which should be sector-specific.

Transport & Environment (T&E) notes that nearly 60 % of the planned battery production in Europe is at risk of not moving forward. This highlights the urgent need for the EU to overhaul its trade and investment policies. T&E estimates that at least €50 billion in public investment will be required to support the EU battery value chain by 2030. T&E also calls for an anti-subsidy investigation into battery cells and packs to enable trade defence measures, noting that tariffs on battery imports are just above 1 %. T&E also calls for a rapid adoption of secondary legislation regarding the carbon footprint of batteries. Clear policy and a long-term vision are key to securing investment in battery supply chains. This includes not only maintaining the 2035 car CO_2 ambition, but committing to additional ambition in electrifying fleets and creating a European compact battery EV industry.

The European Council on Foreign Relations (ECFR) emphasises the need for the EU to drafting a comprehensive economic security doctrine that addresses the geoeconomic landscape. The EU should focus on establishing reliable partnerships to reduce dependencies in sectors where it cannot compete on cost or lacks major industrial players in areas such as anode production. This could be achieved through clean trade and investment partnerships. The EU should also promote 'made-in-EU' measures for battery cells, components and recycling. While this approach would not exclude foreign companies, it would aim to prevent Europe from becoming merely an assembly hub. Under the Batteries Regulation, robust CO_2 footprint disclosure requirements should favour green production in the EU. Furthermore, the EU should consider extending countervailing duties on imports of battery EVs from China to batteries, as a way to incentivise Chinese EV manufacturers in Europe to use locally-made batteries. Furthermore, in areas where the EU battery sector is competitive and could reach technological breakthroughs, the focus should be on scaling commercial production and, if necessary, on imposing tariffs until the industry matures (e.g. solid-state batteries).

Clean Industrial Deal-related policy is expected to be a key element of the EU policy agenda in the next 5 years. In her July 2024 political guidelines for 2024-2029, Commission President Ursula von der Leyen announced a new plan for the EU's prosperity and competitiveness, aimed at making it easier to do business, consolidating the single market, implementing a 'Clean Industrial Deal' – a set of measures to decarbonise and lower energy prices, promote R&I, boost productivity by means of digital technology diffusion – invest massively in the EU's sustainable competitiveness, and tackle the skills and labour gap. In her mission letter to the Commissioner for Prosperity and Industrial Strategy, Stéphane Séjourné, von der Leyen asked him to develop the Clean Industrial Deal within the first 100 days of his term, together with the Commissioner for Climate, Net Zero and Clean Growth. The proposal should focus on decarbonisation, clean technologies and incentives for investment. In particular, it should put forward specific solutions for each of the value chains concerned. Séjourné should also propose an industrial decarbonisation accelerator act to support

the clean tech industry. He will lead the work on the implementation of the CRMA and NZIA. He should also work on a competitiveness coordination tool to improve coordination at all levels. In addition, Séjourné should develop an EU competitiveness fund to channel investment towards innovation and the technologies that will be key for the EU in the future.

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ENDNOTES

[1] This section aims to provide a flavour of the debate and is not intended to be an exhaustive account of all different views on the topic.

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